ALGEBRA 1 · SYSTEMS OF EQUATIONS AND INEQUALITIES

Systems Explorer

Explore & Reflect

Name _____ Date _____ Period _____

Open **Systems Explorer** on Mr. K's Classroom: Algebra 1 → Systems of Equations and Inequalities → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict before you touch the tool

1. Two lines have the same slope but different y-intercepts. How many times will they cross?

2. What does the point where two lines cross tell you?

2 Explore use the tool and record what you see

Equations

Set each system with the sliders.

Line 1	Line 2	What the graph shows	Number of solutions	Solution
$y = x + 1$	$y = -x + 5$			
$y = 2x - 1$	$y = 2x + 3$			
$y = -\frac{1}{2}x + 4$	$y = -\frac{1}{2}x + 4$			
$y = 3x - 2$	$y = x + 2$			

Inequalities: $y \geq x + 1$ and $y \leq -x + 5$

Switch to Inequalities mode and drag the test point.

Test point	Line 1 true?	Line 2 true?	Solution of the system?
(0, 0)			
(1, 4)			
(4, 0)			
(0, 3)			

3 Write the rule fill in each blank from what you observed

1. The solution of a system is the point where the lines _____.

2. Same slope and different y-intercepts: _____ solution.

3. Same slope and same y-intercept: _____ solutions.

4. Different slopes: exactly _____ solution.

5. For a system of inequalities, the solutions are the points where the shadings _____.

4 Check no tool: show your work

1. Solve $y = 2x + 1$ and $y = -x + 7$.

2. How many solutions: $y = 4x - 3$ and $y = 4x + 1$?

3. How many solutions: $y = 3x + 6$ and $y = 3(x + 2)$?

4. Is $(1, 2)$ a solution of $y > x$ and $y \leq 3$?

5 Reflect

How can you tell how many solutions a system has without graphing?